MIL-R-21211A(SHIPS)
6 November 1961
SUPERSEDING
(See section 6)

# MILITARY SPECIFICATION

# RANGES, ELECTRIC, NAVAL SHIPBOARD

## 1. SCOPE

1.1 Scope. - This specification covers standard electric ranges for shipboard use.

1.2 Classification. - Ranges are of the following types as specified (see 6.1). The letters "NM" following the type designation (for example, type A-NM) denote ranges of special nonmagnetic construction (see 3.2.3).

- Type A, Type A-NM } 36 inches wide, consists of:
  - Three 12 by 24 inch combination griddle-hotplates.
  - One 21 inch wide oven.
- Type C, Type C-NM } 30 inches wide, consists of:
  - Two 12 by 24 inch combination griddle-hotplates.
  - One 18 inch wide oven.
- Type S, Type S-NM } 42 inches wide, consists of:
  - One 18 by 21 inch griddle.
  - One 10 by 18 inch combination griddle-hotplate.
  - One 10 by 18 inch hotplate.
  - Two 33 inch wide ovens.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

FSC 7310

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SPECIFICATIONS

MILITARY

MIL-M-14 - Molding Plastics and Molded Plastic Parts, Thermosetting.
MIL-P-116 - Preservation, Method of.
MIL-G-858 - Gray Iron Alloy: Castings (Scale Resisting).
MIL-E-917 - Electric Power Equipment, Basic Requirements for (Naval Shipboard Use).
MIL-D-963 - Drawings, Electrical, Hull and Mechanical Equipment for Naval Shipboard Use.
MIL-P-15037 - Plastic Sheet, Laminated, Thermosetting, Sheets, Glass-Cloth, Melamine-Resin.
MIL-M-15071 - Manual, Service (Instruction Books) for Shipboard Electrical and Mechanical Equipment.
MIL-W-16878 - Wire, Electrical (Insulated, High Temperature).
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.
MIL-H-22577 - Heating Elements, Electrical; Cartridge, Strip and Tubular Type.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-271 - Nondestructive Testing Requirements for Metals.
MS15249 - Fuse, Cartridge, Ferrule Contact, Style F60.
MS15250 - Fuse, Cartridge, Ferrule Contact, Style F61.
MS15251 - Fuse, Cartridge, Ferrule Contact, Style F62.
MS15252 - Fuse, Cartridge, Knife Blade Contact, Style F63.
MS15253 - Fuse, Cartridge, Knife Blade Contact, Style F64.
MS15254 - Fuse, Cartridge, Knife Blade Contact, Style F65.
MS15255 - Fuse, Cartridge, Knife Blade Contact, Style F66.
MS15256 - Fuse, Cartridge, Knife Blade Contact, Style F67.
MS15257 - Fuse, Cartridge, Ferrule Contact, Style F70.
MS15258 - Fuse, Cartridge, Knife Blade Contact, Style F71.



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PUBLICATIONS

BUREAU OF SHIPS
NAVSHIPS 250-692-13 - Radiographic Standards for Steel Castings.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN IRON AND STEEL INSTITUTE
Steel Products Manual.

(Application for copies should be addressed to the American Iron and Steel Institute, 350 5th Avenue, New York 1, N. Y.)

NATIONAL SAFE TRANSIT COMMITTEE, INCORPORATED
National Safe Transit Committee Test Procedures.

(Application for copies should be addressed to the National Safe Transit Committee, Incorporated, Associations Building, 1145 19th Street, N. W., Washington 6, D. C.)

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N. Y.)

3. REQUIREMENTS

3.1 Description. - The range shall consist of a body containing the oven or ovens and a cooking surface, together with a separate control box suitable for bulkhead or overhead mounting.



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3.2 Materials. -

3.2.1 Materials not specified shall be selected on the basis of meeting the performance requirements of the specification at minimum cost.

3.2.2 Where corrosion-resisting metal is specified herein, it shall be corrosion-resisting steel or nickel-copper alloy (monel). Corrosion-resisting-steel shall be AISI type 302 or 304.

3.2.3 Ranges of special nonmagnetic construction shall be constructed entirely of nonmagnetic material unless other material is specifically approved by the bureau or agency concerned. Steel, when used, shall be of a nonmagnetic type. Nonmagnetic material is defined as material which has a magnetic permeability of less than 2.0 after fabrication.

3.2.4 Cadmium, lead arsenic, antimony, and other toxic chemicals shall not be present in any measurable amounts on any surface which may come in contact with food.

3.2.5 Bolts, screws, nuts, or other parts used for securing corrosion-resisting metal parts shall be of corrosion-resisting metal or nickel plated.

3.2.6 The heat-insulating material shall be thoroughly dry rock-wool, glass-wool, or other material satisfactory to the bureau or agency concerned, and shall have a thermal conductivity not greater than 0.28 British thermal unit (B.t.u.) per hour per square foot per 1°F. per inch for a mean temperature of 90°F. when packed to the density used in the equipment (see 3.3.3).

3.2.7 Insulating material used for bases, barriers or other parts of cut-outs, terminal boards, or panels shall be molded material in accordance with Specification MIL-M-14, or sheet material in accordance with type GMG of Specification MIL-P-15037. Insulating panels shall be of ample thickness to insure against breaking or cracking. The use of other insulation, where operating temperatures or other conditions make it necessary, will be permitted with the approval of the bureau or agency concerned.



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3.3 General requirements. -

3.3.1 Ranges shall be suitable for heavy Marine duty, compact, of such rugged construction that minimum repairs will be required when in constant use for long periods, and such that normal repairs or replacements, when necessary, may be easily accomplished by the ship's force.

3.3.2 Ranges shall be so designed that ordinary repairs or replacements may be made from the front without moving the equipment or disturbing adjacent equipment.

3.3.3 The heat-insulating material (see 3.2.6) shall be packed to such a density and in such a manner that it will not settle and leave portions of the wall uninsulated.

3.4 Environmental suitability. - Ranges shall be capable of withstanding normal shipboard vibration and mechanical impact. The range shall not be damaged by the environmental test specified in 4.4.11. Ranges that have passed HI shock tests (see 6.3) or are approved by the bureau or agency concerned as being similar to ranges that have passed HI shock tests are considered to be sufficiently substantial in construction to withstand shipboard environmental conditions without further tests.

3.5 Electrical characteristics. -

3.5.1 The ranges shall be suitable for operation on one of the following voltages as specified (see 6.1).

115 volts direct current (d.c.).
230 volts d.c.
250 (175-355) volts d.c. with supplementary 115 volt, single phase, 60 cycle, alternating current (a.c.) control circuit.
500 (355-710) volts d.c. with supplementary 115 volt, single phase, 60 cycle, a.c. control circuit.
440 volts, 3 phase, 60 cycle a.c.



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3.5.1.1 Ranges rated at 250 volts d.c. shall be capable of satisfactory operation over a voltage range of 175 to 355 volts. At the 355 volt condition, the range shall be suitable for operating continuously for 4 hours. Ranges with this voltage rating are intended for submarine use. The specified kilowatt (kw) input shall be obtained at 250 volts.

3.5.1.2 Ranges rated at 500 volts d.c. shall be capable of satisfactory operation over a voltage range of 355 to 710 volts. At the 710 volt condition, the range shall be suitable for operating continuously for 4 hours. Ranges with this voltage rating are intended for submarine use. The specified (kw) input shall be obtained at 500 volts.

3.5.2 Total kilowatt input shall be within 5 percent (plus or minus) of the following at rated voltage:

Type A - 24.0 kw.
Type C - 15.8 kw.
Type S - 25.6 kw.

3.5.3 All ranges shall be so designed and constructed as to withstand, for a period of 1 minute, a dielectric test voltage of twice the rated voltage plus 1000 volts (see 4.4.8).

3.5.4 The insulation resistance of each range when at normal operating temperatures shall be not less than 1.0 megohm. The insulation resistance when the range is at room temperature shall be not less than 2.0 megohms (see 4.4.9).

3.5.5 The clearance and creepage distances shall be not less those required by Specification MIL-917.

3.5.6 The unbalance between phases of a three-phase range shall not exceed 5.0 kw.

3.5.7 All noncurrent-carrying parts shall be gounded.

3.6 Control devices. - Thermostatic switches, 3-heat switches, relays and contactors shall conform to Specification MIL-E-917.

3.6.1 Thermostatic switches. - Thermostatic switch dials shall be located on the front of the range body. The dial markings shall correspond to the average temperatures (example, 3 for 300) maintained. Each thermostatic switch shall have a positive "OFF" position.



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3.6.2 Three-heat switches. - Three-heat switches for the types A and C ranges shall be located in the separate control box. Three-heat switches for the type S range shall be located on the front of the range body. The switch positions shall be clearly marked "HIGH", "MEDIUM" and "LOW".

3.7 Fuses. - Fuses shall be the type shown on Standards MS15249, MS15250, MS15251, MS15252, MS15253, MS15254, MS15255, MS15256, MS15257, and MS15258. Both sides of each heater circuit shall be protected with fuses suitable for the heater current, but in no case shall the fuse rating be less than 125 percent nor more than 250 percent of the maximum current at rated voltage of the unit protected. Where a control transformer is used, both sides of the transformer primary shall be fused (6 amperes maximum).

3.8 Indicator lights. - An indicator light shall be furnished for each thermostatically controlled surface unit or oven. The light shall be mounted on the range body near the corresponding thermostat dial. The lights shall be illuminated when the heating elements are energized. Indicator lights shall conform with the following description:

Lamp housing - Mounting hole diameter: 11/16 inch
Depth: Not specified; however, sufficient space shall be allowed behind mounting panel to accommodate a replacement housing extending up to 2-1/16 inches behind the front of the panel.
Plastic materials: Suitable for temperature of application.
Resistor: Built-in resistor selected to give high brilliance of lamp whem illuminated.
Lamp - Type NE-51 or NE-51H.
Lens - Color: Red.
Material: Glass or plastic.
Type: Plain or fluted.
Shape: Stovepipe, macrodome or equivalent shape easily viewed from wide angle.

3.9 Heating elements. - Heating elements shall be the tubular type conforming to Specification MIL-H-22577. Terminal connections may be the plug-in type. Heating elements shall be securely fastened in place.



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3.10 Wiring. -

3.10.1 Wiring shall be enclosed, well supported and adequately protected from grease or food spillage and against mechanical damage. Metal edges with which wire may come in contact shall be well rounded and smooth. Wire terminal connections shall be tight. Wiring shall be neatly formed.

3.10.2 Electrical connections within heated zones shall be securely clamped between corrosion-resisting terminal parts. Unit lead wires brought through an oven wall shall be brought through in such a manner that oven vapors or steam will not escape through the lead opening.

3.10.3 Electrical connections shall be of a substantial screw, bolt or plug-type that will not loosen from shipboard environmental conditions or usage. Leads to terminals not easily identified shall be suitably marked for identification.

3.10.4 Wire run in a heated zone exceeding 400°F., or connected to heating unit terminals where the wire terminal connection temperature exceeds 400°F., shall be nickel conductor not smaller than 6530 circular mils (AWG No. 12) with heavy asbestos insulation. If the wire is brazed or welded, the conductor shall be manganese-nickel (AWG No. 12).

3.10.5 Wire carrying heater current, except that covered by 3.10.4, shall be tin coated copper stranded conductor not smaller than 6090 circular mils (approximately AWG No. 12) insulated with silicone rubber with an outer covering of glass braid. The wire shall conform to Type FF of Specification MIL-W-16878.

3.10.6 Wiring for control circtuis shall be tin coated copper stranded conductor not smaller than 1780 circular mils (approximately AWG No. 18) insulated with silicone rubber with an outer covering of glass braid. The wire shall conform to type FF of Specification MIL-W-16878.

3.11 Covers. - Enclosing covers for switches, controls, fuses, and terminals shall require the use of tools for access.



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3.12 Construction. -

3.12.1 Body. - The range body shall be of substantial angle or reinforced sheet metal construction with joints firmly secured to form a sturdy structure and adequately support all parts.

3.12.2 External metal surfaces. - External metal surfaces of the range body, except surface units and underside of the range shall be corrosion-resisting metal.

3.12.3 Range top. - The portion of the range top in front of the surface heating units shall be of corrosion-resisting sheet steel not less 0.078 inch thick. The entire range top shall be suitable reinforced and shall be secured to the range body in such a manner as to insure against warping or buckling when heated

3.12.4 Sectionalized construction (type S range only). - The type S range shall be constructed in sections that are bolted together. The range shall be capable of being dismantled, for installation, so that the largest part of section is not larger than 42 inches wide by 30 inches high by 25 inches deep.

3.12.5 Grease receptacles and drip pans. - The range shall be so constructed that all spilled foods may be easily reached and removed. Ranges shall be designed so that drippage from the top will not run down the range body. Grease from the cooking surface operations shall drain into a large capacity grease receptacle. Drip pans of sufficient size and design to catch spillage shall be provided underneath. The grease receptacles shall have baffles to retard movement of contents. Drip pans and grease receptacles shall be easily removable from the front of the range. Drip pans and grease receptacles shall be provided as follows:

Type A range - Two drip pans and a separate grease receptacle or two drip pans, one of which is designed to serve as a grease receptacle.
Type C range - One drip pan and a separate grease receptacle or two drip pans, one of which is designed to serve as a grease receptacle.
Type S range - Two drip pans and a separate grease receptacle.



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3.12.6 Guard rail. - The range shall be provided with a substantial corrosion-resisting steel guard rail across the full width at the front. The guard rail shall extend beyond the front of the range top from 3 to 4 inches and shall be supported by substantial brackets. The guardrail shall be at approximately the same level as the range top surface.

3.12.7 Sea racks. - A set of substantial steel sea racks shall be provided to hold utensils in place on top of the types A and C ranges and on the griddle-hotplate and hotplate sections of the type S range. The outer bars shall fit into supports mounted near each corner of the range top where utensils are used. Bars shall be interchangeable as far as practicable. Bars shall be removable but suitable provision shall be made for securing the sea rack assembly in the supports to prevent dislodgement under shock. On types A and C ranges, at least two intermediate bars shall be provided in each direction and shall be adjustable to approximately 3 inch spacing intervals. On the type S range, one intermediate bar shall be provided in each direction and shall be adjustable to approximately 2 inch spacing intervals.

3.12.8 Wiring entrance. - Suitable provision shall be made for wiring entrance into the range from the separate control box. This shall consist of a conduit within the range body which shall terminate at the inner end near the connection block and project at the outer end from the top of the range near a rear corner, a sufficient distance to attach a conduit coupling for conduit extension to the control box.

3.12.9 Connection block. - All heater circuit and oven thermostatic switch lead wires within the range shall terminate at a suitable connection block. The connection block shall be so located in the electrical compartment that connections can be easily viewed and serviced from the front of the range, otherwise a removable connection block assembly shall be provided, so that the connection block may be unfastened, pulled out of the electrical compartment for connection of wires, and replaced. Terminals shall be marked for identification. The electrical compartment shall be protected by a removable cover.



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3.13 Separate control box. -

3.13.1 A line connection block and the electrical components such as cutouts with fuses, contactors, control circuit transformer (when furnished) and the three-heat control switches for types A and C ranges shall be mounted in a substantial enclosure designed for overhead or bulkhead mounting. The body of the enclosure shall be of sheet metal not less than 0.0598 inch thick and a removable cover shall be provided. The switch-operating handles shall be located outside the box at the front and shall be clearly identified for the heating units controlled. The box shall be arranged for normal horizontal mounting. Dimensions shall be as shown on figures 1 and 2.

3.13.2 Four tapped holes (1/2-13) in the back and four in the top of the box shall be provided for mounting the control box.

3.14 Surface units. -

3.14.1 General requirements. -

3.14.1.1 Each surface unit shall be a good grade of cast iron, cast steel or heavy sheet steel adequately ribbed or reinforned underneath to minimize warping and with a substantially level, machined and polished cooking surface. The surface units of the types A and C ranges and the griddle-hotplate and hotplate of the type S range shall be provided with a grease seal and thermal insulating spacer between sections. The grease seal shall prevent seepage or leaking of grease between units. The design and materials for the grease seal and insulating spacer shall be such that grease from the cooking surface will not be absorbed.

3.14.1.1.1 Surface units for nonmagnetic ranges shall be either class 1 or class 2 gray iron alloy in accordance with Specification MIL-G-858. Castings shall be annealed at 1750°F. Other material may be used if approved by the bureau or agency concerned.

3.14.1.2 Surface units shall be positively secured in position on the range top to prevent their dislodgement under shock. Suitable means at the front of the range shall be provided for the release and removal of the surface units. Bolt holes will not be permited in the top of the surface units.



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3.14.1.3 Heating elements for the surface units shall be either cast-in as an integral part of the unit or securely fastened to the underside.

3.14.2 Type A range. -

3.14.2.1 The type A range shall be equipped with 3 surface units, each approximately 12 by 24 inches, installed with the long dimension from front to rear of the range. The range top shall be designed to provide grease drainage from the cooking surface into the grease receptacle. Drainage may be accomplished by an adequate grease gutter surrounding the cooking area of each surface unit or by gutters at the sides of the entire cooking surface and troughs at the front and rear.

3.14.2.2 Heating element wattages shall be 6000 watts (±5 percent) for each surface unit.

3.14.2.3 The temperature of each surface unit shall be controlled by a thermostatic switch having a temperature range of 250° to 850°F.

3.14.3 Type C range. -

3.14.3.1 The type C range shall be equipped with 2 surface units, each approximately 12 by 24 inches, installed with the long dimension from front to rear of the range. The range top shall be designed to provide grease drainage from the cooking surface into the grease receptacle. Drainage may be accomplished by an adequate grease gutter surrounding the cooking area of each surface unit or by gutters at the sides of the entire cooking surface and troughs at the front and rear.

3.14.3.2 Heating element wattages shall be 6000 watts (±5 percent) for each surface unit.

3.14.3.3 The temperature of each surface unit shall be controlled by a thermostatic switch having a temperature range of 250° to 850°F.

3.14.4 Type S range. -

3.14.4.1 Arrangement. - Surface units shall be arranged as shown on figure 3.



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3.14.4.2 Griddle. - The left side of the range top shall be a griddle section having a minimum size of 18-1/2 inches wide by 21-1/2 inches deep. The griddle top shall have a minimum thickness of 0.4375 inch and shall be adequately ribbed or reinforced underneath to minimize warping. A splash guard of corrosion-resisting steel shall be provided around the sides and back of the griddle surface. Height of the splash guard shall be 5 inches at the rear and left side of the griddle surface. At the right side, the griddle splash guard shall be tapered from 5 inches at the rear to 2 inches at the front. The front of the left side of the splash guard shall be cut back or slightly rounded at the top. The splash guard shall be formed integrally with or welded in a grease tight manner to the griddle surface.

3.14.4.2.1 The griddle heating elements shall be arranged in two sections, one to heat the left half and the other to heat the right half of the cooking surface. Each half shall be controlled by a separate, indicating, thermostatic switch having a range of 200° to 425°F.

3.14.4.2.2 A grease gutter, the width of the griddle surface, shall be provided at the front of the griddle. The grease gutter shall have a large drain opening at least 4 inches by 1 inch which will drain all grease into the grease receptacle.

3.14.4.2.3 Heating element wattage for the griddle section shall be 8000 watts (±5 percent).

3.14.4.3 Griddle-hotplate. - At the right of the griddle there shall be a griddle-hotplate, approximately 10 inches wide by 18 inches deep, that is suitable as both a griddle and a hotplate. A grease gutter shall be cast or milled into the right side of the griddle-hotplate from front to rear, so that when it is used as a griddle, grease shall not flow onto the adjacent hotplate.

3.14.4.3.1 The griddle-hotplate shall be controlled by an indicating thermostatic switch having a range of 250° to 850°F.

3.14.4.3.2 Heating element wattage for the griddle-hotplate shall be 3800 watts (±5 percent).



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3.14.4.4 Hotplate. - A hotplate unit, the same size as the griddle-hotplate, shall be located at the right of the griddle-hotplate.

3.14.4.4.1 The hotplate shall have the heating units arranged for separate control of the front and rear halves by two 3-heat switches, each providing high, medium and low heats. Wattage at medium and low heats shall be 1/2 and 1/4, respectively, of that at high heat. Switches shall be located on the front of the range body.

3.14.4.4.2 The total heating element wattage shall be 3800 watts (±5 percent) at high heat, 1900 watts for each of the front and rear halves.

3.14.4.5 Drainage. - The assembly consisting of the griddle-hotplate and hotplate shall be surrounded by a drainage trough. The trough in front of this assembly shall contain a large drain opening, at least 4 inches by 1 inch.

3.15 Ovens. -

3.15.1 Type A and type C ranges shall be provided with one oven located within the range body, below the surface units. The type S range shall be provided with two ovens, located one above the other in the range body.

3.15.2 Dimensions. - Minimum inside clear dimensions of the oven compartment with the door closed and all heating elements and decks in place shall be as shown in table I.

Table I - Minimum range oven dimensions.

| Type of range | Height | Width | Depth |
|---|---|---|---|
| | Inches | Inches | Inches |
| A | 12 | 21 | 26 |
| C | 12 | 18 | 21 |
| S | 7-1/2 | 33 | 19-5/8 |



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3.15.3 Lining. - The range ovens shall be substantially constructed with inner linings and heater supports of corrosion-resisting metal. The inner lining shall be not less than 0.0359 inch thick and shall be adequately reinforced and supported in the oven assembly in such a manner as to have a minimum of through-metal to the outer shell. All joints and seams of the lining shall be constructed so as to prevent steam or moisture from entering the heat insulation.

3.15.4 Door. - Each range oven shall be provided with a downward opening, spring-counterbalanced, shelf-type door of substantial construction. All sheet metal parts shall be not less than 0.0478 inch thick. The door shall be mounted on substantial bearings and shall be provided with a corrosion-resisting metal or polished aluminum handle and latch. The door opening shall be the full width and height of the oven compartment. The door, when open, shall rest horizontally and shall be capable of supporting 150 pounds at the outer edge without permanent distortion. In the open position, the interior door panel shall be at the same level as the oven deck. In this position, there shall be an unobstructed opening into the oven of the minimum height and width dimensions specified in 3.15.2.

3.15.5 Thermal insulation. - Each range oven compartment and door shall be well insulated. Insulation shall be such that a maximum input of 1.00 kilowatt-hour per hour is sufficient to maintain each oven, when empty and closed at 425°F. (75°F. room temperature). Minimum thickness of insulation shall be as specified in table II.

Table II - Minimum thickness of insulation.

| Type of range | Sides | Top and bottom | Back | Between ovens | Door |
|---|---|---|---|---|---|
| | Inches | Inches | Inches | Inches | Inches |
| A | 2 | 2 | 2 | - | 1-1/2 |
| C | 2 | 2 | 1-3/8 | - | 1-1/2 |
| S | 1-5/8 | 1-3/4 | 1-1/2 | 2 | 1-1/2 |



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3.15.6 Vent. - Each range oven shall be provided with a suitable vent located at the back of the oven and extending to the top of the range through a flue. All vent and flue parts shall be of corrosion-resisting metal. A suitable damper with convenient means of regulating the damper opening shall be provided. Damper control handles shall be accessible and shall not be obstructed when pots are used on the surface units. The damper control shall have positive open and closed positions. The open and closed positions shall be clearly marked. On the type S range, the controls also shall be marked as being for the upper or lower oven.

3.15.7 Deck. - An oven deck, having approximately the same area as the oven bottom and with raised edges at the back and both sides, shall be provided over the lower heating unit. The deck shall be of corrosion-resisting steel, substantially constructed and reinforced to prevent objectionable warping under heat or damage from severe use. The deck shall have a smooth surface and shall be flush with the bottom of the door opening. Decks shall be removable through the door openings without removing the door or any part of the oven front.

3.15.8 Heating units and controls. - Each oven section shall be provided with two heating units, one located in the top of the compartment and one located in the bottom. Each heating unit shall cover an area approximately equal to that of the oven area and shall be controlled by a 3-heat switch. The heating units shall be so formed and located as to provide a uniform temperature over the oven area at high, medium, or low heat. Wattage input at medium and low heats shall be 1/2 and 1/4 respectively, of that at high heat. The temperature of each oven compartment shall be controlled automatically by an adjustable thermostatic switch having a range of 200° to 425°F. with the thermostat dial located on the front of the range body. Heating units shall be adequately protected against mechanical injury to the elements or connections, and shall be so constructed and assembled that the complete units may be easily removed for repairs or replacement from the front of the oven. Upper heating unit elements shall not extend below the top of the oven opening. The thermostatic switch and actuating element shall be so located in the oven as to be well protected against mechanical injury. The actuating element shall be protected by a perforated or expanded metal shield.



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3.15.9 Wattage. - Oven heating unit wattages shall be within plus or minus 5 percent of the wattages specified in table III.

Table III - Oven heating unit wattages.

| Type of range | Upper heating unit wattage | Lower heating unit wattage |
|---|---|---|
| A | 3000 | 3000 |
| C | 1800 | 1800 |
| S | 2200 | 2800 |

3.16 Holding-down means. - Provision for bolting down the type S range shall be four holes (one near each corner) for 3/4 inch-diameter holding-down bolts in the bottom of the base. Bolt holes shall be accessible through the front of the base. For the types A and C ranges four substantial sheet metal legs, at least 3 inches high and suitably formed and reinforced to provide ample rigidity and prevent objectionable distortion under shock shall be securely welded or bolted to the bottom corners of the range body. A hole shall be provided in the bottom of each leg for a 3/4 inch diameter holding-down bolt.

3.17 Dimensions and weight. - Dimensions and maximum weight shall be as shown on figures 1 and 2.

3.18 Finish. -

3.18.1 Except where corrosion-resisting material is specified herein, ranges and control boxes shall be suitably finished throughout in accordance with the best practice current in the electrical cooking equipment industry. Exterior finish of control box shall be high-grade baked silver-gray, hammer-type enamel.

3.18.2 Where corrosion-resisting metal equipment is specified herein, the exterior surfaces of the range body shall be commercial polish No. 4. All exposed hardware and fittings on corrosion-resisting metal parts shall be heavily nickel plated. Minimum thickness of nickel plating shall be 0.0006 inch.



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3.19 Identification plate. - Each range shall be provided with a corrosion-resisting metal, brass or anodized aluminum identification plate with marking produced by etching, engraving, stamping, casting or photoprocessing. The marking shall be plainly visible, preferably black in color against a light background. The plate shall be securely fastened to the range in a conspicuous location. The plate shall contain the following information:

Electric range, type --- (A, C or S, as applicable).
Name of manufacturer.
Manufacturer's type or catalog number.
Serial number,
Wattage.
Voltage.
Phase.
Frequency.
Manufacturer's drawing number.
Federal stock number (if available).
Contract or order number.
Year of manufacture.
Component identification number (if available).

3.20 Performance. -

3.20.1 Thermostatically controlled surface units. -

3.20.1.1 Temperature distribution. - Temperature distribution shall be practically uniform over the entire cooking surface. At the peak of the temperature cycle, the temperature at each of the 12 points selected in accordance with figure 4 shall be within 60 degrees (plus or minus) of the average of the 12 temperatures, for a thermostat dial setting of 425°F.

3.20.1.2 Temperature control and thermostat calibration. -

3.20.1.2.1 Definitions. - The following definitions shall apply:

(a) Average cooking surface temperature. - The average of the maximum and minimum cooking surface temperatures measured in three consecutive cycles after steady surface conditions have been reached.



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(b) Maximum and minimum cooking surface temperatures. - The average of the 12 temperatures, at locations shown on figure 4, measured simultaneously at the peaks and valleys, respectively, of the temperature cycle.

(c) Steady surface condition. - The condition reached after the surface heating unit has been under the control of the thermostat for the first four cycles of operation of the thermostat or the condition reached after the thermostat cycling period becomes uniform, whichever occurs last.

(d) Temperature amplitude. - The average difference between maximum and mimimum cooking surface temperature measured in 3 consecutive cycles after steady surface condition has been reached.

3.20.1.2.2 Requirements. - Temperature control shall be such that when checked at thermostat dial settings of 325°F., 375°F., 425°F. and 750°F. (where this setting is provided) the following conditions shall be met at each dial setting:

(a) Average cooking surface temperature (see 3.20.1.2.1(a)). - The average cooking surface temperature shall be within 5°F. (plus or minus) of the dial setting, except at the 750°F. setting where it shall be within 15°F. (plus or minus).

(b) Temperature amplitude (see 3.20.1.2.1 (d)). - The temperature amplitude shall not exceed 75°F., except at the 750°F. setting. Temperature amplitude at the 750°F. setting is not limited.

3.20.1.3 Heating time. - The surface units, except the type S range griddle, shall be capable of reaching a cooking surface temperature (average of the 12 temperature measurements) of 750°F. within 30 minutes, starting at room temperature (approximately 75°F.). The type S range griddle shall be capable of reaching a cooking surface temperature of 400°F. within 12 minutes starting at room temperature (approximately 75°F.).

3.20.2 Hotplate - (type S range). -

3.20.2.1 Temperature distribution. - With both halves of the hotplate operating on high heat, the temperature distribution shall be such that after 30 minutes of operation, the temperature of any of 8 points selected at random shall not vary by more than 150°F. from the average of the 8 temperatures.



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3.20.2.2 Heating time. - With both halves of the hotplate operating on high heat, a temperature of 750°F. (average of temperatures measured at 8 points selected at random on hotplate surface) shall be reached within 30 minutes starting at room temperature, (approximately 75°F.).

3.20.3 Ovens. -

3.20.3.1 Temperature distribution. - Temperature distribution shall be practically uniform throughout the oven. At the peak of the temperature cycle, the temperature of each of the 9 points selected in accordance with figure 5 shall be within 25°F. (plus or minus) of the average of the nine temperatures for a thermostat dial setting of 425°F.

3.20.3.2 Temperature control and thermostat calibration. -

3.20.3.2.1 Definitions. - The following definitions shall apply:

(a) Average oven temperature. - The average of the maximum and minimum oven temperatures measured in three consecutive cycles after steady oven conditions have been reached.
(b) Maximum and minimum oven temperature. - The average of the nine temperatures, at locations shown on figure 5, measured simultaneously at the peaks and valleys, respectively, of the temperature cycle.
(c) Steady oven condition. - The condition reached after the oven has been under control of the thermostat for the first four cycles of operation of the thermostat or the condition reached after the thermostat cycling period becomes uniform, whichever occurs last.
(d) Temperature amplitude. - The average difference between maximum and minimum oven temperature measured in 3 consecutive cycles after steady oven condition has been reached.

3.20.3.2.2 Requirements. - Temperature control shall be such that when checked at thermostat dial settings of 325°F., 375°F. and 425°F., the following conditions shall be met at each dial setting:

(a) Average oven temperature (see 3.20.3.2.1 (a)). - The average oven temperature shall be within 5°F. (plus or minus) of the thermostat dial setting.
(b) Temperature amplitude (see 3.20.3.2.1.(d)). - The temperature amplitude shall not exceed 30°F.



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3.20.3.3 Heating time. - The ovens shall be capable of reaching a temperature (average of the 9 temperature measurements) of 400°F. within 22 minutes starting at room temperature (approximately 75°F.) when empty and closed with both switches at the high heat setting.

3.21 Repair parts and tools. -

3.21.1 Each repair part shall be identical to the originally installed part and shall be suitable for immediate use in replacing the originally installed part in any identical range. In general, the design shall be such that no special tools are required; however, where the design and construction are such as to require the use of certain special tools for their proper maintenance, such tools shall be furnished. Special tools are defined as those tools not listed in the Navy Stock List of General Stores. (Copies of this stock list may be consulted in the office of the Government inspector.)

3.21.2 Onboard repair parts furnished on a ship basis. - When repair parts are consolidated and furnished on a ship basis, the quantities shall be based on the toal number of ranges installed. Table IV shall be used to determine the quantities of parts to be furnished.

Table IV - Onboard repair parts.

| Part | FACTOR | FOR QUANTITY OF ONBOARD REPAIR PARTS REQUIRED PER SHIP<br>Multiply the allowance factor in the applicable column X the quantity installed in one range and round the product to the nearest whole number.<br>EXAMPLE: .10 = 1, 1.4 = 1, 1.5 = 2<br>NUMBER OF RANGES INSTALLED PER SHIP | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | 1 | 2 | 3 | 4 | 5-8 | 9-20 | 21-50 | 51-100 |
| Heating elements, each size and type | R | .20 | .40 | .60 | .60 | .80 | 1.50 | 3.00 | 3.50 |
| Switches, complete | T | .05 | .10 | .15 | .15 | .20 | .25 | .70 | 1.50 |
| Relays or contactors, complete[1] | R | .20 | .40 | .60 | .60 | .80 | 1.50 | 3.00 | 3.50 |
| Thermostatic switches, complete | S | .10 | .20 | .30 | .30 | .40 | .60 | 1.50 | 2.50 |
| Transformer, if used | T | .05 | .10 | .15 | .15 | .20 | .25 | .70 | 1.50 |

[1]One set of contacts and springs and one coil and resistor of each type, as applicable, shall be considered the equivalent of one complete relay or contactor in the furnishing of repair parts.



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3.21.3 Repair parts for stock equipment. - Repair parts in accordance with table IV shall be furnished with ranges purchased for stock. The quantity of parts shall be based on one range being installed per ship.

3.22 Drawings. - Drawings conforming to class A of Specification MIL-D-963 shall be furnished for ranges. In addition to the requirements of Specification MIL-D-963, drawings shall include the following minimum data:

(a) Front view, side view, top view and sectional view, as necessary, with mounting dimensions, overall dimensions, and principal parts indicated by numbered arrows.
(b) Table of important data including: principal cooking surface and oven dimensions, watts input on the various heats, time to reach operating temperature, energy to maintain temperature, net weight, manufacturer's type designation, voltage, wattage.
(c) Wiring diagram.
(d) Internal connections of switches.
(e) Table of phase balance.
(f) Detail of identification plate.
(g) Elementary schematic diagram.

3.23 Service manuals. - Service manuals conforming to Specification MIL-M-15071 shall be furnished.

3.24 When any specification which forms a part of this specification requires that the product be subjected to and pass qualification tests, only products which have been approved for listing on the applicable Qualified Products List on the date of invitation for bids or date of initiation of purchase action shall be utilized. In the event no Qualified Products List has been issued, the contractor shall request instructions as to what testing will be required to determine whether the product meets the requirement of the specification.

3.25 Workmanship. - Workmanship shall be first class and in accordance with the best commercial practice.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Govenment as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling for acceptance inspection. -

4.2.1 Inspection lot. - All ranges of the same type, design and voltage offered for delivery at one time shall be considered a lot for purpose of acceptance inspection.

4.2.2 Sampling for group A tests. - A random sample of ranges shall be selected from each lot in accordance with table V and shall be subjected to each of the group A tests specified in 4.3.1 with lot acceptance based on the sampling requirements in accordance with Standard MIL-STD-105.

Table V - Sampling for group A tests
AQL (approx.) = 1.5 percent defective.

| Number of ranges in inspection lot | Number of ranges in sample | Number of ranges non-conforming on any group A test | |
|---|---|---|---|
| | | Acceptance number | Rejection number |
| 15 and under | 7 | 0 | 1 |
| 16 to 40 | 10 | 0 | 1 |
| 41 to 110 | 15 | 0 | 1 |
| 111 to 300 | 25 | 1 | 2 |
| 301 to 500 | 35 | 1 | 2 |
| 501 and over | 50 | 2 | 3 |



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4.2.3 Sampling for group B tests. - A random sample of ranges shall be selected from each lot in accordance with table VI and shall be subjected to each of the group B tests specified in 4.3.2.

Table VI - Sampling for group B tests.

| Number of ranges in inspection lot | Number of ranges in sample | Number of ranges non-conforming on any group B test | |
|---|---|---|---|
| | | Acceptance number | Rejection number |
| 15 and under | 3 | 0 | 1 |
| 16 to 40 | 5 | 0 | 1 |
| 41 to 110 | 7 | 0 | 1 |
| 111 to 300 | 10 | 0 | 1 |
| 301 and over | 15 | 1 | 2 |

4.2.4 Sampling for group C tests. - The first range of a particular type, design and voltage shall be subjected to the group C tests specified in 4.3.3. On subsequent contracts or orders, tests on ranges of identical construction need not be repeated, provided that a copy of the original test report is available. The environmental tests of 4.4.11 conducted on one range will be considered applicable to all ranges of that type, regardless of voltage, provided that the basic design and construction are the same.

4.2.5 Sampling for inspection of grey iron alloy surface unit castings conforming to Specification MIL-G-858 shall be as specified in 4.3.5.



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4.3 Acceptance inspection. -

4.3.1 Group A tests. - Each of the sample ranges selected in accordance with 4.2.2 shall be subjected to each of the tests specified herein, and the results of each test compared with the requirements of this specification. Failure to conform to this specification for any group A tests shall be counted as a defect and the range shall be rejected. If the number of such nonconforming equipments in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

| Group A tests | Reference |
|---|---|
| Adjustment, fit and materials | 4.4.1 |
| Dimensions | 4.4.2 |
| Wattage | 4.4.3 |
| Dielectric (cold) | 4.4.8 |
| Insulation resistance (cold) | 4.4.9 |

4.3.2 Group B tests. - Each of the sample ranges selected in accordance with 4.2.3 shall be subjected to each of the tests specified herein, and the results of each test compared with the requirements of this specification. Failure to conform to this specification for any group B tests shall be counted as a defect and the range shall be rejected. If the number of such nonconforming equipments in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

| Group B tests | Reference |
|---|---|
| Operation | 4.4.4 |
| Dielectric (hot) | 4.4.8 |
| Insulation resistance (hot) | 4.4.9 |

4.3.3 Group C tests. - Each of the sample ranges selected in accordance with 4.2.4 shall be subjected to each of the tests specified herein, and the results of each test compared with the requirments of this specification. Ranges shall not be accepted for delivery until these tests have been satisfactorily completed. Failure of the range to pass any of the tests shall result in the rejection of all units so manufactured until the cause of failure has been corrected on all ranges and the sample has been successfully retested. Ranges of the same type, voltage and design need not be retested on subsequent contracts or orders; therefore, a record of the test shall be retained by the manufacturer.



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| Group C tests | Reference |
|---|---|
| Performance | 4.4.5 |
| Energy to maintain oven temperature | 4.4.6 |
| High voltage (d.c. submarine ranges only) | 4.4.7 |
| Creepage and clearance | 4.4.10 |
| Environmental | 4.4.11 |

4.3.4 Repair parts. - All repair parts shall be examined to determine that the parts are duplicates of the parts furnished in the ranges. If the inspector has any reason to doubt the ready interchangeability of the repair parts with the original parts, he may require a demonstration of interchangeability. The wattage of heating elements and calibration of thermostatic switches furnished as repair parts shall be measured in accordance with the sampling for group A and group B tests, respectively (see 4.2.2 and 4.2.3).

4.3.5 Inspection of gray iron alloy surface unit castings. - Gray iron alloy surface unit castings conforming to Specification MIL-G-858 shall be inspected by dye-penetrant and radiographic examination in accordance with Standard MIL-STD-271 in addition to the inspection required by Specification MIL-G-858. All castings shall meet the requirements of class II of NAVSHIPS 250-692-13. Radiographic inspection is modified to the following:

(a) Radiographic inspection at inspection level I of Standard MIL-STD-105. Acceptable quality level 1.5 percent.
(b) Dye-penetrant inspection of all castings at all fillet areas only.

4.4 Test procedures. - All operating tests shall be conducted at rated voltage. Where temperature measurements are required, thermocouples shall be used.

4.4.1 Adjustment, fit and materials. -

4.4.1.1 Each of the sample ranges shall be subjected to a thorough examination to ascertain that the materials, workmanship and design are in strict conformance with this specifcation and approved drawing. The fit of parts shall be observed with particular reference to the interchangeability of parts as are likely to require replacement during the normal service life of the equipment, and also with reference to ease of removal and replacement of such parts as drip-pans and sea racks, and their satisfactoriness with respect to their intended purpose. The weight of the range and control box shall be taken and recorded on the test report.



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4.4.2 Dimensions. - Each of the sample ranges shall be measured to determine conformance with the dimensions shown on figures 1 and 2.

4.4.3 Wattage. - The wattage of each range section and heating elements furnished as repair parts shall be measured and compared with the wattage requirements as tabulated in table VII.

Table VII - Tabulation of wattage requirements.

| Unit | Wattage (±5 percent) | | |
|---|---|---|---|
| | High | Medium | Low |
| Type A range | | | |
| Griddle-hotplate | 6000 | ---- | --- |
| Oven, upper unit | 3000 | 1500 | 750 |
| Oven, lower unit | 3000 | 1500 | 750 |
| Type C range | | | |
| Griddle-hotplate | 6000 | ---- | --- |
| Oven, upper unit | 1800 | 900 | 450 |
| Oven, lower unit | 1800 | 900 | 450 |
| Type S range | | | |
| Griddle | 8000 | ---- | --- |
| Griddle-hotplate | 3800 | ---- | --- |
| Hotplate (each half) | 1900 | 950 | 475 |
| Oven, upper unit | 2200 | 1100 | 550 |
| Oven, lower unit | 2800 | 1400 | 700 |

4.4.4 Operation. - The purpose of this test is to determine if the range is functioning properly and is equivalent in performance to the range having passed the performance test (see 4.4.5).



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4.4.4.1 Thermostatically controlled surface units. -

4.4.4.1.1 At thermostat dial settings of 375°F. for all thermostatically controlled surface units, the surface temperature directly above each thermostat sensing element shall be measured at the peaks and valleys of 3 consecutive temperature cycles after the steady surface condition (see 3.20.1.2.1 (c)) has been reached. From this data, the following shall be determined:

(a) Thermostat calibration. - Each surface unit shall be checked to ascertain that its thermostat is properly calibrated for average cooking surface temperature (see 3.20.1.2.1 (a)). The average temperature above the thermostat sensing element shall be calculated by averaging the 6 temperatures measured (3 peaks and 3 valleys) and compared with the average temperature at the same point determined in the performance test (see 4.4.5.1.1). The difference between the average temperature determined in this test and the average temperature determined in the performance test shall not exceed 5°F.

(b) Temperature amplitude. - The temperature amplitude, determined by calculating the average temperature difference between the 3 peaks and 3 valleys, shall be within 5°F. (plus or minus) of that measured at the same point in the performance test (see 4.4.5.1.1).

4.4.4.1.2 Temperature distribution. - At a thermostat dial setting of 425°F., the temperature at 5 points on the suface of each unit shall be measured at the peak of the temperature cycle, after the steady surface condition has been reached. Two of the points shall be within 1 inch of the edge of the surface. The remaining 3 points shall be selected at random. None of the temperatures measured shall vary by more than 60°F. (plus or minus) from maximum cooking surface temperature determined in the performance test (see 4.4.5.1.1).

4.4.4.2 Hotplate (type S range). - Switches controlling each half of the hotplate shall be operated to determine that they are functioning properly. Wattage at high, medium and low heats on each half of the hotplate shall be measured to verify that the hotplate and switches are connected properly.



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4.4.4.3 Ovens. - At a thermostat dial setting of 375°F., the temperature at the center point (see figure 5) shall be measured at the peaks and valleys of 3 consecutive temperature cycles after the steady oven condition (see 3.20.3.2.1 (c)) has been reached. From this data, the following shall be determined:

(a) Thermostat calibration. - Each oven shall be checked to determine that its thermostat is properly calibrated for average oven temperature (see 3.20.3.2.1 (a)). The average temperature at the center point shall be calculated by averaging the 6 temperatures measured (3 peaks and 3 valleys) and compared with the average temperature at the same point determined in the performance test (see 4.4.5.3.1). The difference between the average temperature determined in this test and the average temperature determined in the performance test shall not exceed 5°F.

(b) Temperature amplitude. - The temperature amplitude, determined by calculating the average temperature difference between the 3 peaks and 3 valleys, shall be within 5°F. (plus or minus) of that measured at the same point in the performance test (see 4.4.5.3.1).

4.4.5 Performance. -

4.4.5.1 Thermostatically controlled surface units. - Temperatures shall be measured at 12 points on each surface unit, selected as shown on figure 4. Adjacent surface units shall be operated with the same temperature settings as the unit under test and may be tested concurrently if desired. Temperature measurements shall be recorded to determine conformance of the surface units with the following requirements of 3.20.1:

(a) Temperature distribution (see 3.20.1.1). - With a 425°F. thermostat setting, temperature shall be measured at peak of temperature cycle.

(b) Temperature control and thermostat calibration (see 3.20.1.2). - At each thermostat setting, 325°F., 375°F., 425°F. and 750°F. (where this setting is provided), temperatures shall be measured at the peaks and valleys of 3 consecutive temperature cycles after the steady surface condition has been reached.

(c) Heating time (see 3.20.1.3). - At the maximum thermostat setting, temperatures shall be measured to determine if the specified temperature is reached within the specified time.



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4.4.5.1.1 Additional data. - For future reference in the comparisons required for the operation test (see 4.4.4.1), the average thermostat temperature at a dial setting of 375°F. shall be determined by averaging the peaks and valleys for 3 consecutive temperature cycles at a point on the surface directly above the thermostat sensing element. The temperature amplitude at this point shall be determined by calculating the average difference between the 3 peaks and 3 valleys. Also, the maximun cooking surface temperature (see 3.20.1.2.1 (b)) at a dial setting of 425°F. shall be calculated and recorded.

4.4.5.2 Hotplate (type S range). - Temperatures shall be measured at 8 points selected at random on the hotplate surface after the hotplate has been operating for 30 minutes at high heat. The temperature distribution and heating time shall be checked for conformance with 3.20.2.

4.4.5.3 Ovens. - Temperatures shall be measured at 9 points in the oven compartment, selected as shown on figure 5. Temperature measurements shall be recorded to determine conformance of the oven with the following requirements of 3.20.3:

(a) Temperature distribution (see 3.20.3.1). - With a 425°F. thermostat setting, temperatures shall be measured at peak of temperature cycle.
(b) Temperature control and thermostat calibration (see 3.20.3.2). - At each thermostat setting, 325°F., 375°F. and 425°F., temperatures shall be measured at the peaks and valleys of 3 consecutive temperature cycles after steady oven temperature has been reached.
(c) Heating time (see 3.20.3.3). - With a 425°F. thermostat setting, temperatures shall be measured to determine if the specified temperature is reached within the specified time.

4.4.5.3.1 Additional data. - For future reference in comparisons required for the operation test (see 4.4.4.3), the average temperature at the center point (see figure 5) of the oven compartment at a thermostat dial setting of 375°F. shall be determined by averaging the peaks and valleys for 3 consecutive temperature cycles after steady oven temperature has been reached. The temperature amplitude at the center point shall be determined by calculating the average difference between the 3 peaks and 3 valleys.



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4.4.6 Energy to maintain oven temperature. - With the oven empty and closed and under the control of the thermostatic switch set at 425°F., the energy required to maintain that temperature shall be measured to determine conformance with 3.15.5. The switches controlling the upper and lower heating units shall both be set at high heat.

4.4.7 High voltage (d.c. submarine ranges only). - Ranges rated at 250 (175 - 355) volts d.c. or 500 (355 - 710) volts d.c. shall be operated continuously at their maximum voltage of 355 or 710 volts, respectively, for a period of at least 4 hours to determine conformance with 3.5.1.

4.4.8 Dielectric. - The range shall be subjected to and shall withstand, for a period of 1 minute, a dielectric test voltage of twice the maximum rated voltage plus 1000 volts applied between each circuit and the body of the equipment. During the test all other circuits and metal parts shall be effectively grounded and all relays or contactors shall be closed. Dielectric tests shall, in general, be made upon the completely assembled equipment and not upon individual parts. The dielectric test (cold) shall be conducted at normal room temperature. The dielectric test (hot) shall be conducted after all sections of the range have been operating on high heat for a period of 45 minutes.

4.4.9 Insulation resistance. -

4.4.9.1 The insulation resistance test shall be made with all circuits of equal voltage above ground connected together. Circuits or groups of circuits of different voltages above ground shall be tested separately. Tests shall be made using a potential of 500 volts d.c.

4.4.9.2 Each range shall have insulation resistance measurements made to determine conformance with 3.5.4.

4.4.9.3 The insulation resistance test shall be made under the following conditions:

(a) Insulation resistance (hot) - As soon as practicable following the dielectric test (hot).
(b) Insulation resistance (cold) - At approximately room temperature.



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4.4.10 Clearance and creepage. - Clearance and creepage distances shall be demonstrated by actual measurements to be in accordance with Specification MIL-E-917.

4.4.11 Environmental test. - Before packaging, the range shall be subjected to the following vibration and impact tests using test equipment in accordance with National Safe Transit Committee standards:

(a) Vibration test. - The equipment to be tested shall be placed on the table of the vibration tester. Fences, suitable for the equipment being tested may be attached to the test table. Vibration frequency shall be determined by advancing the cycle of vibration until the equipment under test leaves the table sufficiently to allow a thin piece of cardboard, approximately 1/16 inch thick, to be inserted between the bottom edge of the equipment and the platform of the machine. The equipment shall be vibrated for 2 hours.

(b) Impact test. - An incline impact testing device of the Conbur type, or other testing machine producing equivalent results, and a shock recorder shall be used. The equipment to be tested shall be securely fastened on the dolly in such a position that the equipment will make no physical contact with the barrier of the impact machine. The shock recorder shall be securely fastened on the equipment to record the maximum shock received during the impact test. The dolly shall be drawn up to incline to a position which will produce an impact into at least the first quarter of the fifth zone and then released. The test shall be repeated so that the equipment receives two impacts in each horizontal plane.

Failure to operate satisfactorily at the conclusion of the tests or evidence of breakage or disassembly shall be cause for rejection.

4.5 Inspection of preservation, packaging and packing. - Inspection of preservation, packaging and packing to verify conformance with the requirements of section 5 shall be performed in accordance with the sampling plans and inspection procedures of Specification MIL-P-116.



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## 5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts. -

5.1.1. Ranges. -

5.1.1.1 Preservation and packaging. - Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.1.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature. Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.1.2 Onboard repair parts. - Onboard repair parts shall be preserved and packaged level A, packed level C, and marked levels A and C respectively in accordance with the item specification or Specification MIL-E-17555, as applicable. Items not covered therein shall be unit protected and packaged in accordance with the guide lines of Specification MIL-P-116. Items shall be individually packaged except when used in sets.

5.2 Domestic shipment and storage or overseas shipment. - The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).



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(5.2.1 The following provides the various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

5.2.1.1 Preservation and packaging. -

5.2.1.1.1 Ranges. -

5.2.1.1.1.1 Level A. - Extent of disassembly of the ranges shall be in accordance with the recommendations of the equipment manufacturer, subject to the approval of the Government inspector. Ranges shall be individually preserved and packaged in accordance with method I of Specification MIL-P-116. Surfaces of the range which come in contact with food shall be coated with preservative compound conforming to type P-14. Preserved surfaces shall be covered or wrapped with barrier material conforming to grade A of Specification MIL-B-121 and secured with water resistant paper tape. Thermostat and switch knobs, and all other projections shall be adequately cushioned to prevent damage from handling, shipping and storage.

5.2.1.1.2 Repair parts. - Repair parts shall be preserved and packaged level A in accordance with Specification MIL-E-17555 or the item specification, as applicable. Items not covered therein shall be unit protected and packaged in accordance with the guide lines of Specification MIL-P-116. Parts shall be individually packaged except when used in sets.

5.2.1.2 Packing. -

5.2.1.2.1 Ranges. -

5.2.1.2.1.1 Level A. - Ranges shall be packed for overseas shipment in nailed wood or wood-cleated plywood boxes conforming to Specification PPP-B-621 (class 2), or PPP-B-601 (overseas type), respectively, at the option of the contractor. Boxes shall be modified by the addition of nominal 2 by 4 inch skids as specified in the applicable box specification. Shipping containers shall have case liners conforming to Specification MIL-L-10547. Case liners shall be sealed and closed in accordance with the appendix to Specification MIL-L-10547. The box shall be closed and strapped as specified in the applicable box specification or appendix thereto.



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5.2.1.2.1.2 Level B. - Ranges shall be packed for domestic shipment in domestic type nailed wood or wood-cleated plywood boxes conforming to Specification PPP-B-621 (class 1) or PPP-B-601 (domestic type), respectively, at the option of the contractor. Box closure shall be as specified in the applicable box specification or appendix thereto.

5.2.1.2.1.3 Anchoring, blocking, bracing and cushioning shall be in accordance with Specification JAN-P-100.

5.2.1.2.2 Repair parts. - Repair parts shall be packed level A or B as specified, in accordance with Specification MIL-E-17555.

5.2.1.3 Service manuals. - Service manuals shall be packaged, packed and marked for the level specified in accordance with Specification MIL-M-15071.

5.2.1.4 Marking. - In addition to any special marking required, interior packages and exterior shipping containers shall be marked for shipment in accordance with Standard MIL-STD-129 and the applicable packaging specification. )

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of range (see 1.2).
(c) Electrical characteristics (see 3.5.1).
(d) Length of wire for connections to separate control box if special length is desired (see 3.10.7).
(e) Preservation, packaging, packing or marking requirements other than those required by 5.1 (see 5.2).

6.2 Supersession data. - This specification supersedes Specification MIL-R-21211(SHIPS) dated 10 April 1958, and that part of Specification MIL-C-001218B(SHIPS) dated 30 September 1957 covering "Ranges" and used in lieu of Specification MIL-C-1218 dated 15 July 1949.



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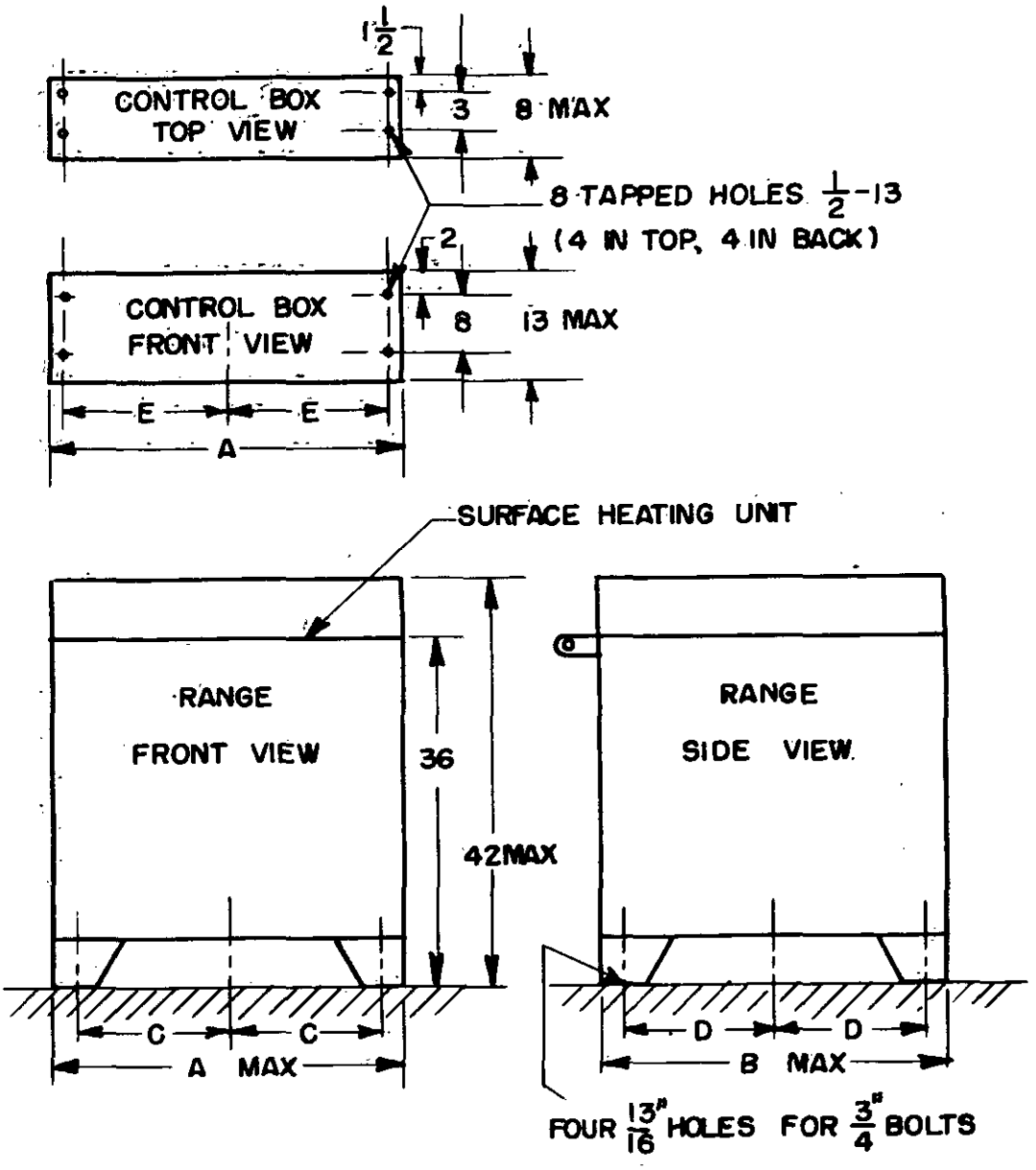
6.3 Ranges that have passed the HI shock test that was formerly required for ranges under Specification MIL-C-001218B(SHIPS) or ranges that are approved by the bureau or agency concerned as being similar in construction to ranges that have passed the HI shock tests, are considered to meet the requirments of 3.4 without being tested as described in 4.4.11.

Notice. - When Government drawings, specification, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specification, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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| TYPE | A | B | C | D | E | WEIGHT |
|---|---|---|---|---|---|---|
| A | 36 | 36 1/4 | 16 3/4 | 16 3/4 | 17 7/16 | 830 |
| C | 30 | 30 1/4 | 13 11/16 | 12 1/8 | 14 7/16 | 550 |

NOTES: 1. DIMENSIONS IN INCHES.

SW4610

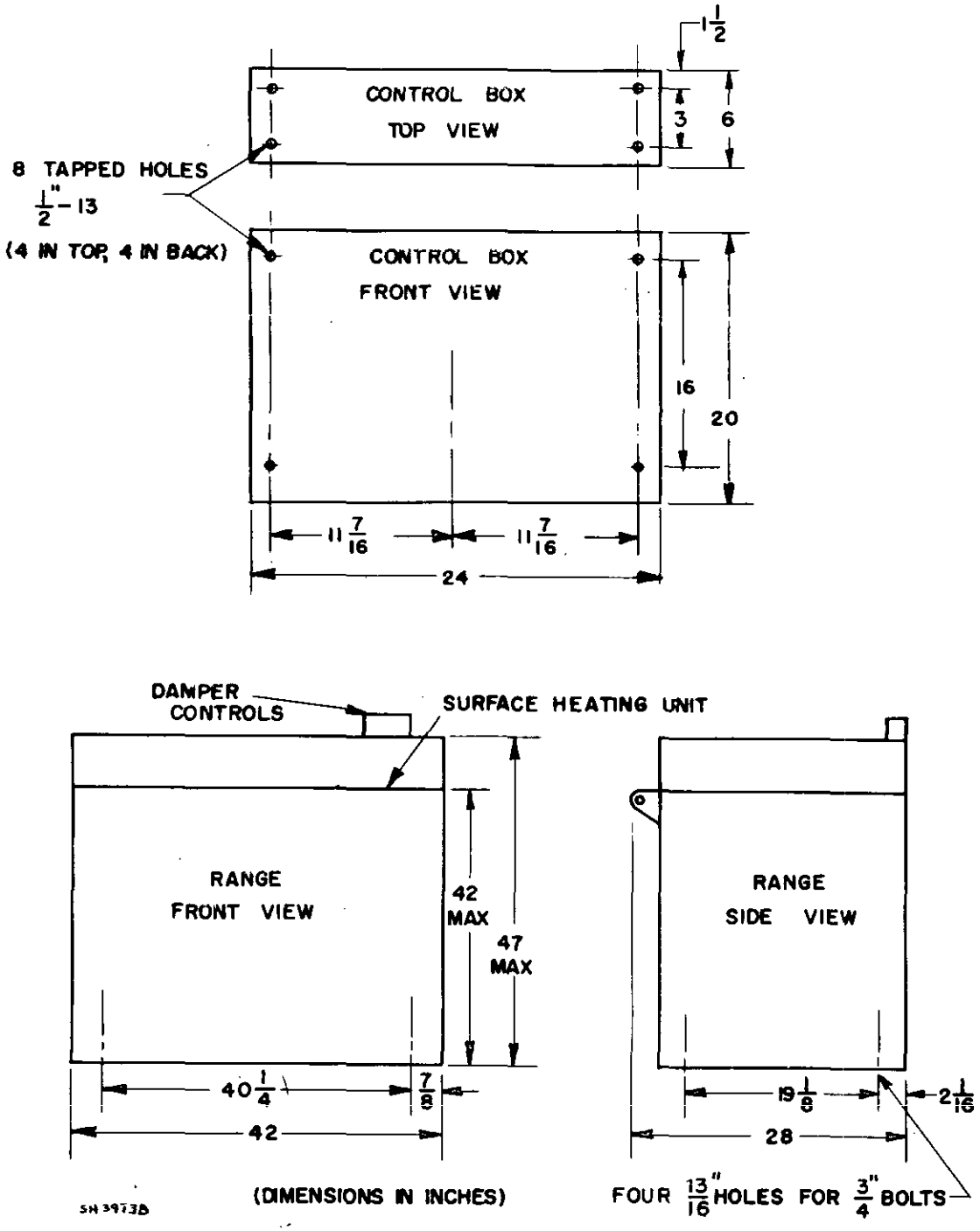
2. TOLERANCE: ± 1/16 INCH

3. WEIGHT IS MAXIMUM COMBINED WEIGHT OF RANGE AND CONTROL BOX.

Figure 1 - Dimensions and weights of type A and C ranges.



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TOLERANCE: $\pm \frac{1}{16}$

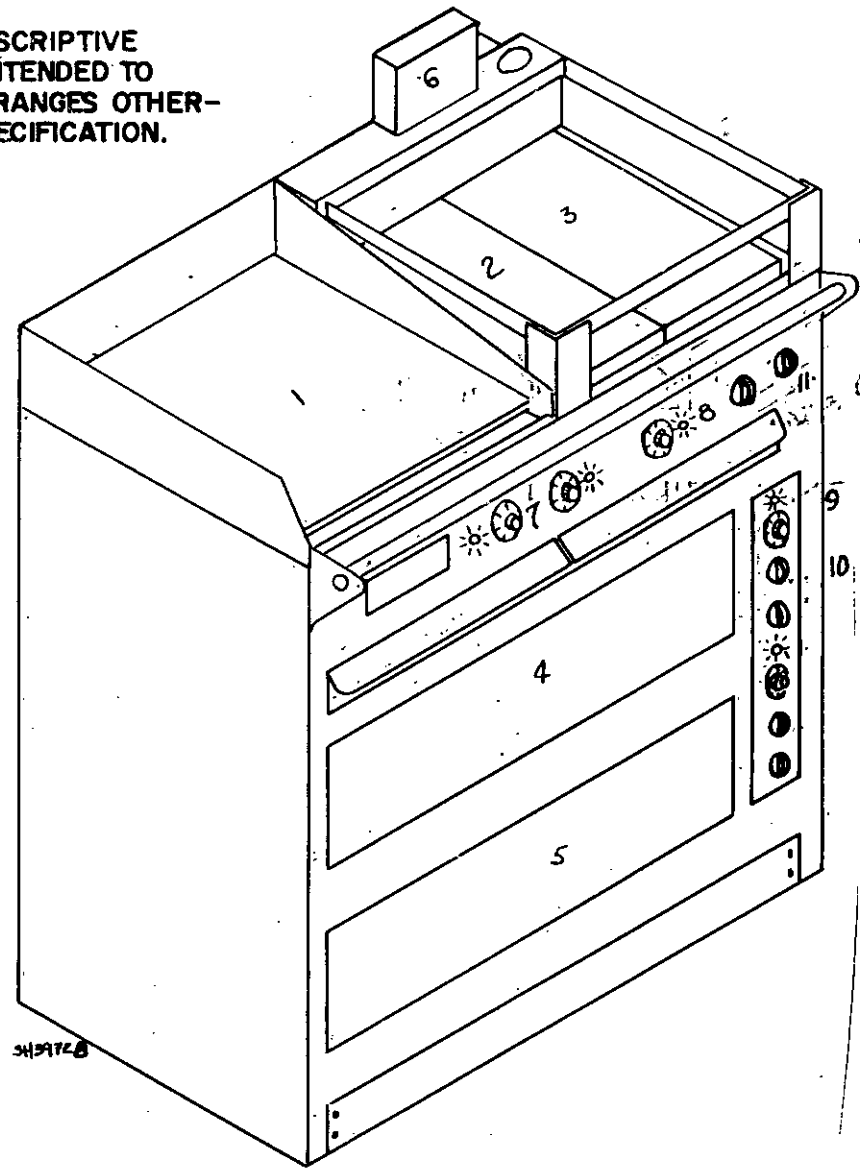
WEIGHT: 875 POUNDS MAX. (INCLUDES CONTROL BOX)

Figure 2 - Dimensions and weight of type S range.



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THIS ILLUSTRATION IS FOR DESCRIPTIVE PURPOSES ONLY AND IS NOT INTENDED TO PRECLUDE THE PURCHASE OF RANGES OTHERWISE CONFORMING TO THIS SPECIFICATION.



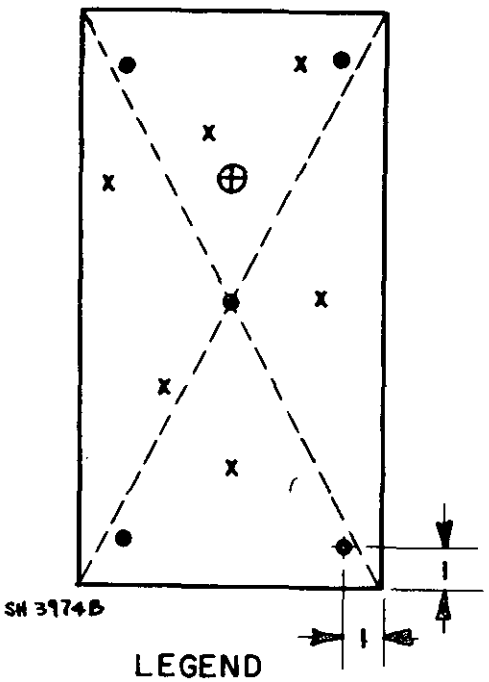

1. GRIDDLE
2. GRIDDLE-HOTPLATE
3. HOTPLATE
4. UPPER OVEN
5. LOWER OVEN
6. OVEN DAMPER CONTRULS
7. GRIDDLE THERMOSTATS
8. GRIDDLE-HOTPLATE THERMOSTAT
9. OVEN THERMOSTATS
10. OVEN SWITCHES
11. HOTPLATE SWITCHES

Figure 3 - Type S range illustration.



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LOCATION OF TEMPERATURE MEASUREMENT POINTS ON THERMOSTATICALLY CONTROLLED SURFACE UNITS FOR PERFORMANCE TEST.



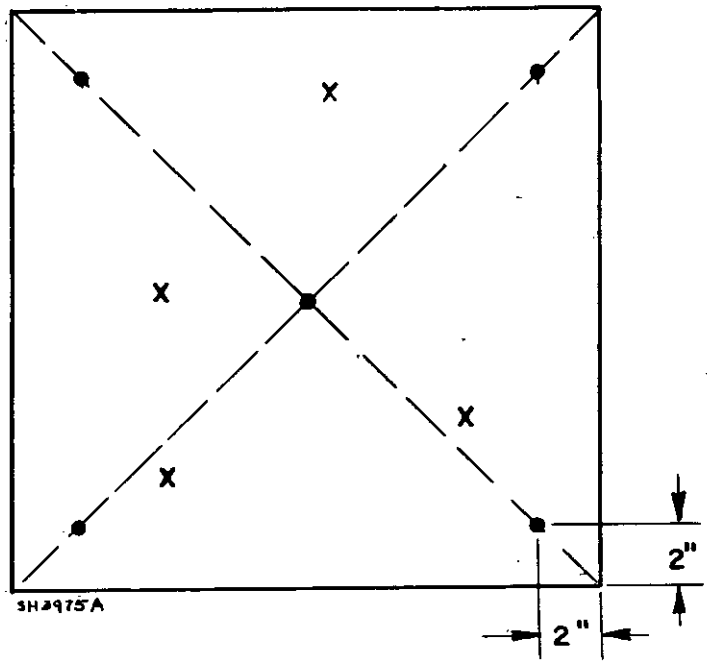

4 CORNER POINTS (1 INCH FROM EDGES)
1 CENTER POINT (INTERSECTION OF DIAGONALS)
1 POINT ABOVE THERMOSTAT SENSING ELEMENT
6 POINTS SELECTED AT RANDOM WITH 1 POINT IN EACH SECTION BOUNDED BY DIAGONALS.

Figure 4 - Thermostatically controlled surface unit temperature measurement points.



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LOCATION OF TEMPERATURE MEASUREMENT POINTS FOR OVEN PERFORMANCE TEST.




THE FOLLOWING POINTS LOCATED IN A HORIZONTAL PLANE 4 INCHES ABOVE THE OVEN DECK:

4 CORNER POINTS (2 INCHES FROM EDGES)
1 CENTER POINT (INTERSECTION OF DIAGONALS)
4 POINTS SELECTED AT RANDOM WITH 1 POINT IN EACH SECTION BOUNDED BY DIAGONALS.

LEGEND ● FIXED POINTS
X RANDOM POINTS

Figure 5 - Oven temperature measurement points.



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